

004.01 Data Lake Cleansing Policy

 Page Status - Ready for Review

Description

This page is designed to define the data cleansing policy and its implementation process.

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What is a Cleansing Process?

On a high level, cleansing is the process:

- Taking the raw data from the source system
- Applying an expected schema
- Verifying whether the data confirms to the schema
- Applying standardized and special cleansing rules to:
 - Fix bad data
 - Quarantine bad data

Data Cleansing Policy Requirements

- Identification of Primary Key
- Identification of delete and replace keys
- Load at Same Grain as Source at Raw and Cleansed
- Trim all values from source
- Fix Bad Text Patterns
- Schema drift
- Record Flag Status (bad data)
- Partition of load file
- Validity of reference or orphan columns
- Retention of special characters where applicable

- Conversion of date identified columns to UTF-8
- Conversion of numeric columns
- Conversion of numeric to identified negative or positive
- Carriage and Line Feed Return removal
- Default value assignment
- Identification of duplicates
- Null or required field identification
- Conversion to required format (phone #, postal code)
- Addition of provenance fields
 - job_id
 - batch_id
 - audit_create_date
 - audit_change_date
 - audit_create_user
 - audit_change_user
 - source_file_name
 - partition_ts
 - bad_record_data
 - bad_record_flag
 - bad_record_type

Characteristics of Data Cleansing Playbook

- Apply Schema:
 - Apply a predefined schema onto the raw data by creating external tables on the raw file location
- Perform Schema Validation
 - Validate whether the raw data confirms to the specified schema and mark the record as bad data if it doesn't conform to the schema.
- Apply Data cleansing Rules
 - Apply configurable data cleansing rules on the raw data. For e.g. Trim, Padding the data etc.
- Data Type Harmonization
 - Apply transformations to the source data in order to convert to the standard data type. This includes parsing dates, numbers etc. from various formats to their respective standard data

types

- Default Value Assignment
 - Assign configurable values to missing data as per the datatypes.
- Mandatory Columns Validation
 - Validate that each records have all of the mandatory columns. Flag a row as a bad record if any of the mandatory columns are missing.
- Remove Bad characters
 - Remove any configure characters from the incoming data fields.
- Data Deduplication
 - Validate the primary keys of the data and if duplicate records are identified then apply rules to select the winning record and mark the other as bad data.